ALGEBRA 2 · UNIT 9: CONIC SECTIONS

Signals & Dishes

Use circles, parabolas, ellipses, and hyperbolas to design communication technology.

CP1

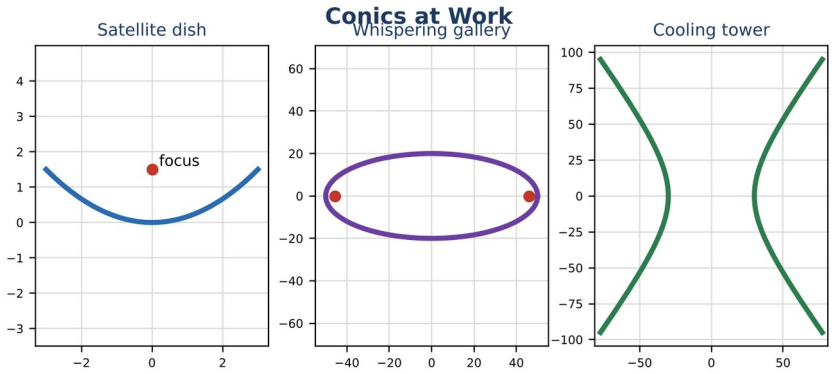
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

SignalWorks builds cell towers, satellite dishes, and even whispering galleries for science museums. Each design depends on one of the conic sections.

You will analyze each design with its equation, then create a coverage plan for a new town.



Learning Goals

- I can write conic equations in standard form by completing the square.
- I can find foci, vertices, and asymptotes.
- I can design with conic properties.

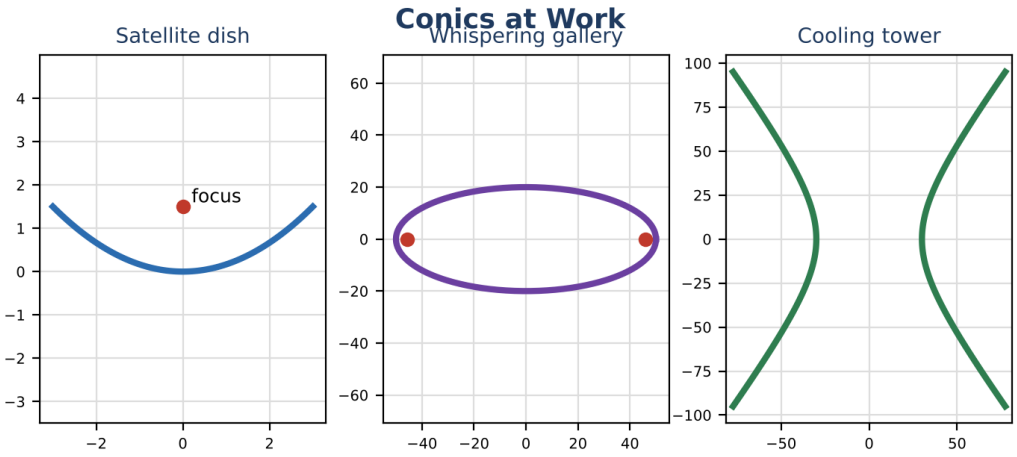
Project Timeline

When	What you will do
Day 1	Towers & Dishes
Day 2	Ellipses & Hyperbolas
Day 3	Coverage Map
Final	Engineering Design Brief + reflection

Part 1: Towers & Dishes

Day 1 · Write each equation in standard form.

1. A tower's coverage boundary is $x^2 + y^2 - 6x + 2y - 6 = 0$. Complete the square to find its center and radius.



2. Design a dish that is 8 ft across and 1 ft deep. Write its equation $x^2 = 4py$ and find the focus.

3. Write the equation of a circle centered at $(-2, 4)$ that passes through $(1, 8)$.

Part 2: Ellipses & Hyperbolas

Day 2 · Find all key features.

4. For $x^2 \div 2500 + y^2 \div 400 = 1$, find the vertices, co-vertices, and foci.

5. For $x^2 \div 900 - y^2 \div 1600 = 1$, find the vertices, foci, and asymptotes.

6. Classify each: $4x^2 + 9y^2 = 36$; $x^2 - y^2 = 16$; $y = 2x^2 - 3$; $x^2 + y^2 = 25$.

Part 3: Coverage Map

Day 3 · Cover houses (0, 0), (4, 2), (−3, 5), and (6, −4) with as few 5-mile towers as possible.

7. Place the towers and write their equations.

8. Show every house is covered.

9. Explain why your number of towers is the minimum (or near it).

Final Product: Engineering Design Brief

Your final product must include:

- Standard-form equations
- Dish design
- Feature tables
- Coverage plan

Reflection

R1. Why does completing the square reveal a circle's center?

R2. How do ellipses and hyperbolas differ in the relationship among a, b, and c?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Coverage plan uses correct circle equations and point tests, and every conic feature is correct. Goes beyond the minimum with a creative, realistic choice.	Coverage plan uses correct circle equations and point tests, and every conic feature is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Polished, organized, and visually engaging.	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.